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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 19

Design a C program to implement process synchronization using mutex locks.

PROGRAM:

```
#include <stdio.h>
#include <pthread.h>

int counter = 0;
pthread_mutex_t mutex;

void *threadFunction(void *arg)
{
    for (int i = 0; i < 1000000; i++)
    {
        pthread_mutex_lock(&mutex);
        counter++;
        pthread_mutex_unlock(&mutex);
    }
    return NULL;
}

int main()
{
    pthread_t thread1, thread2;

    pthread_mutex_init(&mutex, NULL);

    pthread_create(&thread1, NULL, threadFunction, NULL);
    pthread_create(&thread2, NULL, threadFunction, NULL);

    pthread_join(thread1, NULL);
```

```
    pthread_join(thread2, NULL);

    printf("Final counter value: %d\n", counter);

    pthread_mutex_destroy(&mutex);

    return 0;
}
```

OUTPUT:

